Student 23 **MOHAMED IMRAAN S** 192424001RA 2 / 2 submitted

Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Main{
public static void main(String[]args){
    Scanner S1=new Scanner(System.in);

    double Foodcost=S1.nextDouble();
    S1.nextLine();
    String Coupon=S1.nextLine();

    Foodorder f=new Foodorder(Foodcost,Coupon);
    f.displayDetails();
}
}
interface DeliveryCharge{
    void calculateDeliveryCharge();
}
interface CouponDiscount{
    void calculateDiscount();
}
class Foodorder implements DeliveryCharge,CouponDiscount{
    double Foodcost;
    String Coupon;
    double DeliveryCharge;
    double Discount;
    double FinalBill;

    Foodorder(double Foodcost,String Coupon){
        this.Foodcost=Foodcost;
        this.Coupon=Coupon;
    }
    public void calculateDeliveryCharge(){
        if(Foodcost>=500)
            DeliveryCharge=0;
        else
            DeliveryCharge=50;
    }
    public void calculateDiscount(){
        if(Coupon.equals("SAVE10"))
            Discount=Foodcost*0.10;
        else
            Discount=0;

        FinalBill=Foodcost+DeliveryCharge-Discount;
    }
    void displayDetails(){
        calculateDeliveryCharge();
        calculateDiscount();

        System.out.println("Food Cost : "+Foodcost);
        System.out.println("Delivery Charge : "+DeliveryCharge);
        System.out.println("Discount : "+Discount);
        System.out.println("Final Bill : "+FinalBill);
    }


}
```

INPUT

```
600
SAVE10
```

OUTPUT

```
Food Cost: 600.0
Delivery Charge: 0.0
Discount: 60.0
Final Bill: 540.0
```

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[]args){
        Scanner S1=new Scanner(System.in);
        int roomNumber=S1.nextInt();
        S1.nextLine();
        String guestName=S1.nextLine();
        double roomRent=S1.nextDouble();

        HotelBill h=new HotelBill(roomNumber,guestName,roomRent);
        h.calculateServiceCharge();
        h.displayDetails();
    }
}
interface ServiceCharge{
void calculateServiceCharge();
}
class Room{
    int roomNumber;
    String guestName;
    double roomRent;

    Room(int roomNumber,String guestName,double roomRent){
        this.roomNumber=roomNumber;
        this.guestName=guestName;
        this.roomRent=roomRent;
    }
}
class HotelBill extends Room implements ServiceCharge{
    double serviceCharge;
    double finalBill;

    HotelBill(int roomNumber,String guestName,double roomRent){
        super(roomNumber,guestName,roomRent);
    }
    public void calculateServiceCharge(){
        if(roomRent>=5000)
            serviceCharge=roomRent*0.12;
        else
            serviceCharge=roomRent*0.08;

        finalBill=roomRent+serviceCharge;
    }
    void displayDetails(){
        System.out.println("Room Number : "+roomNumber);
        System.out.println("Guest Name : "+guestName);
        System.out.println("Room Rent : "+roomRent);
        System.out.println("Service Charge : "+serviceCharge);
        System.out.println("Final Bill : "+finalBill);
    }
}
```

INPUT

```
205
Ananya
6000
```

OUTPUT

```
Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0
```